



CHAPTER

3

Classification of Elements and Periodicity in Properties

Sr. No.	Concept	Formulas	Other information
1.	Dobereiner's triads	Atomic mass of 2 nd element $= \frac{\text{Atomic mass of 1st element} + \text{Atomic mass of 3rd element}}{2}$	Triad is a group of three elements arranged in increasing order of atomic mass.
2.	Screening effect / Shielding constant (σ): (a) For s and p electrons (b) For d and f electrons	$[(\text{Number of electrons in } n^{\text{th}} \text{ shell}) - 1] \times 0.35 + \text{number of electrons in } (n-1) \text{ shell} \times 0.85 + \text{number of electrons in } (n-2) \text{ shell or all inner shells} \times 1]$ $[(\text{Number of electrons in } n^{\text{th}} \text{ shell}) - 1] \times 0.35 + (\text{number of electrons in all inner shells}) \times 1]$	N.A.
3.	Effective nuclear charge (Z_{eff})	$Z_{\text{eff}} = Z - \sigma$	Here, Z is atomic number and σ is shielding constant
4.	Atomic size	Number of shells $\propto$ Atomic size $\text{Atomic size} \propto \frac{1}{Z_{\text{eff}}}$ $\text{Atomic size} \propto \frac{1}{\text{Magnitude of +ve charge}}$ $\text{Atomic size} \propto \text{Magnitude of -ve charge}$	Here, Effective nuclear charge (Z_{eff})

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5.	Ionisation energy (I.E.)	$\text{I.E.} \propto Z_{\text{eff}}$ $\text{I.E.} \propto \frac{1}{\text{Atomic size}}$ $\text{I.E.} \propto \text{Magnitude of +ve charge}$ $\text{I.E.} \propto \frac{1}{\text{Magnitude of -ve charge}}$ $\text{I.E.} \propto \frac{1}{\text{Metallic character}}$ $\text{I.E.} \propto \frac{1}{\text{Reactivity of metal}}$	Here, Effective nuclear charge (Z_{eff})
6.	Electron affinity (E.A.)	$\text{E.A.} \propto \frac{1}{\text{Atomic size}}$ $\text{E.A.} \propto \frac{1}{\text{Screening effect}}$ $\text{E.A.} \propto Z_{\text{eff}}$	Here, Effective nuclear charge (Z_{eff})
7.	Electronegativity (E.N.):	$\text{E.N.} \propto Z_{\text{eff}}$ $\text{E.N.} \propto \frac{1}{\text{Atomic size}}$ $\Delta \text{E.N.} \propto \text{Non-metallic nature}$ $\Delta \text{E.N.} \propto \frac{1}{\text{Metallic nature}}$	Here, Effective nuclear charge (Z_{eff}) and $\Delta \text{E.N.}$ is electronegativity difference.
	(a) Pauling's scale	$X_A - X_B = 0.208 \sqrt{E_{A-B} - \sqrt{E_{A-A} \times E_{B-B}}}$ (All bond energies are in kcal/mol) $X_A - X_B = 0.102 \sqrt{E_{A-B} - \sqrt{E_{A-A} \times E_{B-B}}}$ (All bond energies are in kJ/mol)	X_m is electronegativity on Mulliken's scale. E_{A-B} is the bond energy of A–B. E_{A-A} is the bond energy of A–A. E_{B-B} is the bond energy of B–B. X_p is electronegativity on Pauling scale.
	(b) Mulliken's scale	$X_m = \frac{\text{I.E.} + \text{E.A.}}{2} \text{ eV/atom}$ $2X_m - \text{I.E.} - \text{E.A.} = 0$	I.E. is ionisation energy. E.A. is electron affinity.
	(c) Allred Rochow's scale	$X = 0.744 + \frac{0.359 Z_{\text{eff}}}{r^2}$	
	(d) Relation between Pauling and Mulliken scale of electronegativity	$X_p = \frac{X_m}{2.8}$	

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8.	Vanderwaal radii (VWR)	$VWR = 2 \times C.R.$	Here, C.R. is covalent radii
9.	Identification of group, period and block: (a) Period number (b) Block (c) Group number For s-block elements For p-block elements For d-block elements For f-block elements	Period number = Principal quantum number of valence shell electron in ground state electronic configuration. (Highest value of n) Block = The last electron enters in which subshell. Group number = the number of valence shell electrons Group number = 10 + number of valence shell electrons Group number = number of electrons in (n – 1) d sub shell + number of electrons in valence shell Group number = 3	N.A.